

Nausea and Vomiting in Adults

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Nausea is an unpleasant feeling of needing to vomit. People also may feel dizziness, vague discomfort in the abdomen, and an unwillingness to eat.

Vomiting is a forceful contraction of the stomach that propels its contents up the esophagus and out the mouth. (See also [Vomiting in Infants and Children](#).) Vomiting empties the stomach and often makes people with nausea feel considerably better, at least temporarily. Vomiting is quite uncomfortable and can be violent. Severe vomiting can project stomach contents many feet (projectile vomiting).

Vomiting is not the same as [regurgitation](#), which is the spitting up of stomach contents without forceful abdominal contractions or nausea. For instance, people with [achalasia](#) or [Zenker diverticulum](#) may regurgitate undigested food without nausea.

Vomit^{us}—the material that is vomited up—usually reflects what was recently eaten. Sometimes it contains chunks of food. When blood is vomited, the vomitus is usually red ([hematemesis](#)), but if the blood has been partly digested, the vomitus looks like coffee grounds. When bile is present, the vomitus is bitter and yellow-green.

Complications

In addition to being uncomfortable, vomiting can cause complications:

- Inhaled vomitus (aspiration)
- Torn esophagus ([Mallory-Weiss syndrome](#), [esophageal rupture](#))
- [Dehydration](#) and [electrolyte abnormalities](#)
- [Undernutrition](#) and weight loss

People who are unconscious or only partly conscious can inhale their vomitus. The acid in the vomitus can severely irritate the lungs.

Vomiting greatly increases pressure within the esophagus, and severe vomiting can tear the lining of the esophagus (see [Mallory-Weiss Syndrome](#)). A small tear causes pain and sometimes bleeding, but a large tear can be fatal.

Because people lose water and minerals (electrolytes) in vomitus, severe vomiting can cause [dehydration](#) and [electrolyte abnormalities](#). Newborns and infants are particularly susceptible to these complications.

Chronic vomiting can cause [undernutrition](#), weight loss, and metabolic abnormalities.

Causes of Nausea and Vomiting

Nausea and vomiting result when the vomiting center in the brain is activated. Causes typically involve disorders of the digestive tract or the brain, or ingested substances.

Common causes

The most common causes of nausea and vomiting are:

- Gastroenteritis (infection of the digestive tract)
- Substances (such as medications or illicit drugs)
- Toxins

Nausea and vomiting commonly occur with any dysfunction of the digestive tract but are particularly common with [gastroenteritis](#). A less common digestive tract disorder is [obstruction of the intestine](#), which causes vomiting because food and fluids back up into the stomach because of the obstruction. Many other abdominal disorders that cause vomiting also cause significant abdominal pain. In such disorders (for example, [appendicitis](#) or [pancreatitis](#)), it is typically the pain rather than the vomiting that causes people to seek medical care.

Many substances, including alcohol, opioid analgesics (such as [morphine](#)), cannabis (marijuana), and chemotherapy agents, can cause nausea and vomiting.

Toxins, such as [lead](#) or those found in some foods and plants, can cause severe nausea and vomiting.

Less common causes

Less common causes of nausea and vomiting include:

- Brain or central nervous system disorders

- Motion sickness
- Metabolic changes or bodywide (systemic) illness
- Psychological disorders
- Chronic nausea and vomiting syndrome
- Cyclic vomiting syndrome

The vomiting center in the brain also can be activated by certain brain or central nervous system disorders, including infections (such as [meningitis](#) and [encephalitis](#)), [migraines](#), and disorders that increase pressure inside the skull (intracranial pressure). Disorders that increase intracranial pressure include [brain tumors](#), [brain hemorrhage](#), and severe [head injuries](#).

The balance organs of the inner ear (vestibular apparatus) are connected to the vomiting center. This connection is why some people become nauseated by the movement of a boat, car, or airplane ([motion sickness](#)) and by certain disorders of the inner ear (such as [labyrinthitis](#) and [positional vertigo](#)).

Nausea and vomiting may also occur when there are metabolic changes in the body, such as during early pregnancy, or when people have very poorly controlled [diabetes](#) or severe [liver failure](#) or [kidney failure](#).

Psychological problems also can cause nausea and vomiting. Such vomiting may be intentional. For instance, people who have [bulimia](#) make themselves vomit to lose weight. Or it may be unintentional. For instance, children who are afraid of going to school vomit as a response to their psychological distress.

Chronic nausea and vomiting syndrome is a centrally mediated syndrome or disorder of the gut-brain axis. That is, the nausea and vomiting are not caused by a physical or specific psychological disorder but rather may be a problem with the way the brain and gastrointestinal system communicate. In this syndrome, people have bothersome nausea, vomiting, or both that occur at least once a week for at least 6 months including the last 3 months and that have no identifiable cause after testing has been done.

Cyclic vomiting syndrome is an uncommon disorder in which people have severe attacks of vomiting (or sometimes only nausea) at varying intervals. People feel normal between attacks. Although it usually starts in childhood, it sometimes lasts until adulthood. Cyclic vomiting that begins in adulthood is often due to chronic marijuana (cannabis) use ([cannabinoid hyperemesis syndrome](#)). The vomiting can be relieved by a hot shower and goes away after people stop using marijuana.

Evaluation of Nausea and Vomiting

Not every episode of nausea and vomiting requires immediate evaluation by a doctor. The following information can help people decide whether a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

Certain symptoms and characteristics are cause for concern. They include

- Signs of [dehydration](#) (such as thirst, dry mouth, little or no urine output, and feeling weak and tired)
- Headache, stiff neck, confusion, or decreased alertness
- Constant abdominal pain
- Tenderness when the abdomen is touched
- Distended (swollen) abdomen

When to see a doctor

People who have warning signs should see a doctor right away, as should people who vomited any blood or who recently had a head injury.

People who have nausea and vomiting but no warning signs should see a doctor if vomiting continues for more than 24 to 48 hours or if they are unable to tolerate more than a few sips of liquid. People who have a few episodes of vomiting (with or without diarrhea) but are able to tolerate at least some liquids should call their doctor. Depending on their age, other symptoms, and known medical conditions (such as cancer or diabetes), the doctor may suggest that people be seen for an evaluation or stay home and try simple remedies.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the vomiting and the tests that may need to be done (see table [Some Causes and Features of Nausea and Vomiting](#))

During the history, doctors ask whether the person is pregnant or has diabetes, migraines, liver or kidney disease, or cancer (including the timing of any chemotherapy or radiation therapy). All recently ingested medications and substances

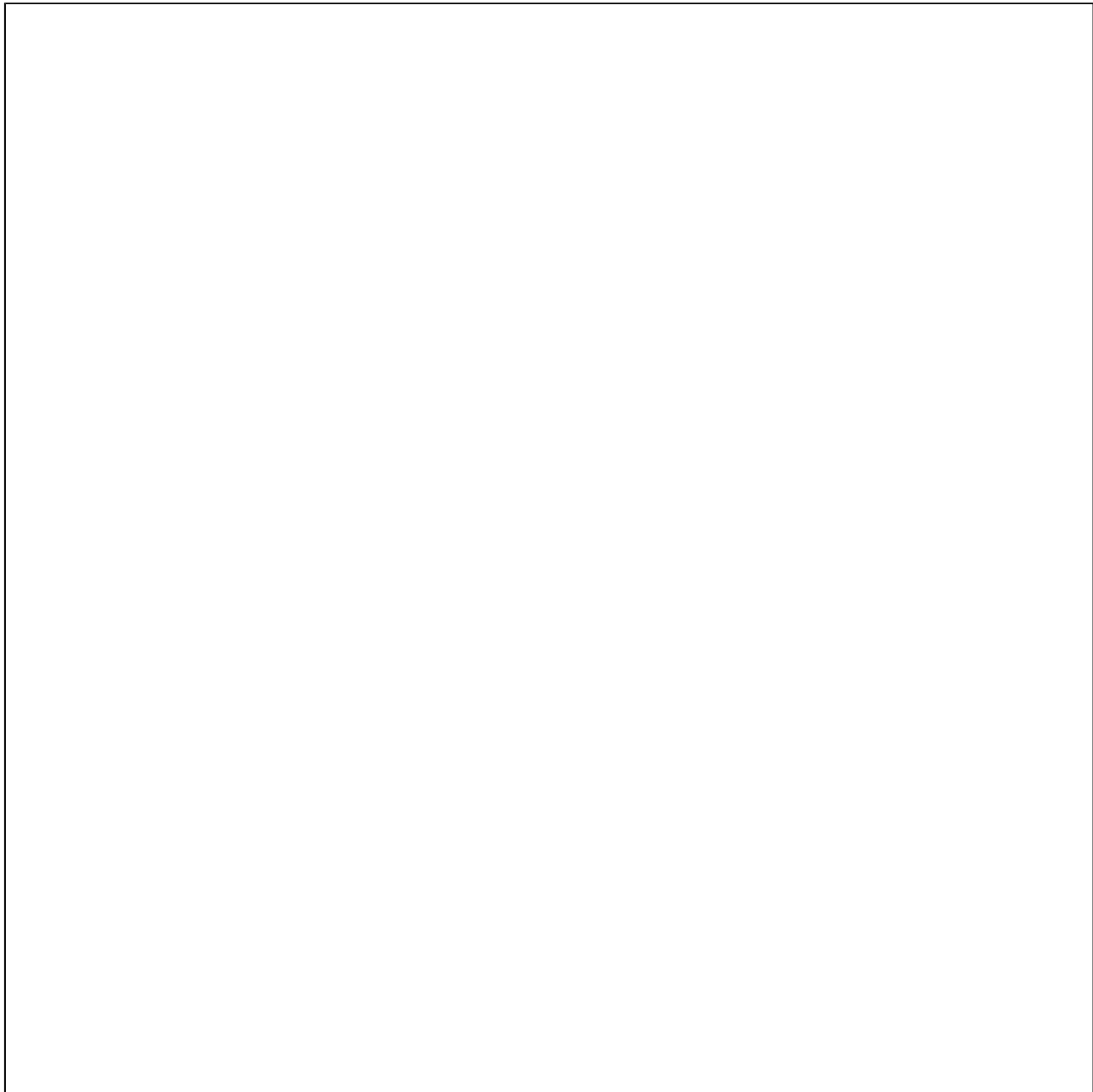
are noted because certain substances may not be toxic until several days after ingestion (such as acetaminophen and some mushrooms).

During the physical examination, doctors look for the following:

- Signs of dehydration (such as a rapid heart rate, low blood pressure, and dry mouth)
- Signs of a serious abdominal disorder (such as a distended abdomen and/or severe tenderness to the touch)
- Decreased alertness or any other neurologic abnormalities suggesting a brain disorder

Doctors note any previous abdominal surgery because fibrous bands of scar tissue (adhesions) may have formed and caused an intestinal obstruction.

Although people with previously known disorders that cause vomiting (such as migraine) may simply be having a recurrence of that disorder, doctors thoroughly look for signs of a new, different problem.



Some Causes and Features of Nausea and Vomiting

Cause	Common Features*	Tests†
Digestive tract disorders		
<u>Appendicitis</u> or another sudden, severe disorder within the abdomen (such as a <u>perforated intestine</u> , <u>gallbladder inflammation</u> , or <u>pancreatitis</u>)	Significant abdominal pain Abdomen that is tender to the touch	Abdominal imaging tests (such as x-rays, <u>ultrasound</u> , and/or <u>CT</u>)
<u>Intestinal obstruction</u>	No bowel movements and no flatus Cramping abdominal pain that comes and goes Distended abdomen Usually in people who are known to have a hernia or who have had abdominal surgery	Abdominal x-rays taken with the person in flat and upright positions
<u>Gastroenteritis</u>	Vomiting and diarrhea Little or no abdominal pain (except during vomiting) Barely fever, or blood in stool	A doctor's examination
Gastroparesis (poor stomach emptying)	Vomiting of partially digested food a few hours after ingestion	A doctor's examination Abdominal x-rays taken when the



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